

Numerical Methods for PDEs

by Andrea Cangiani

International School for Advanced Studies (SISSA)

September 2026

Contents

1	Well-posed problems and reliable discretizations	1
1.1	Why numerical methods?	1
1.2	Discrete problems	1
2	Elliptic tools: Green identities and maximum principles	4
2.1	Gauss–Green formulas	4
2.2	Harmonic functions	4
2.3	Poisson’s equation: uniqueness and stability	4
2.4	A general second-order operator	5
3	Centered finite differences in one dimension	7
3.1	The model problem and its discretization	7
3.2	The discrete maximum principle	7
3.3	A sharp stability estimate	8
3.4	Consistency and convergence	8
3.5	Conditioning and solver cost	9
4	Weak formulations and energy principles	10
4.1	Why the strong formulation is too restrictive	10
4.2	Sobolev spaces	10
4.3	Derivation of the weak problem	11
4.4	Dirichlet’s principle	11
5	Finite elements and finite volumes in one dimension	13
5.1	The conforming Galerkin idea	13
5.2	Well-posedness and error analysis	13
5.3	When finite elements reproduce finite differences	14
5.4	A finite volume derivation	14

1 Well-posed problems and reliable discretizations

1.1 Why numerical methods?

Analytical theory answers structural questions: Does a solution exist? Is it unique? How does it depend on the data? Even when those questions have satisfactory answers, a closed formula is rarely available. Moreover, the solution belongs to an infinite-dimensional function space. A numerical method replaces the continuous problem by a finite-dimensional problem whose solution can be computed.

Write an abstract problem as

$$\mathcal{P}(u, g) = 0, \quad (1.1)$$

where g collects the forcing, coefficients, domain, and boundary data. When the solution is unique, the solution operator is denoted by $u = R(g)$.

Definition 1.1 (Hadamard well-posedness). Problem (1.1) is well posed in specified data and solution norms if:

1. a solution exists for every admissible datum;
2. the solution is unique;
3. the solution depends continuously on the data.

In a locally Lipschitz setting, continuous dependence takes the form

$$\|R(g + \delta g) - R(g)\|_X \leq C_g \|\delta g\|_Y$$

for all sufficiently small admissible perturbations δg . For nonzero g and $R(g)$, a local relative condition number is

$$\kappa_R(g) = \limsup_{\varepsilon \downarrow 0} \sup_{0 < \|\delta g\|_Y \leq \varepsilon \|g\|_Y} \frac{\|R(g + \delta g) - R(g)\|_X / \|R(g)\|_X}{\|\delta g\|_Y / \|g\|_Y}.$$

A large condition number signals intrinsic sensitivity. *No numerical algorithm can remove sensitivity that already belongs to the continuous problem.*

1.2 Discrete problems

Let $h > 0$ denote a resolution parameter, usually a mesh width. For each h , let X_h be the finite-dimensional space of discrete unknowns and let Y_h be the space in which the discrete equations, residuals, and right-hand sides are measured. Both spaces are equipped with suitable norms. Once the discrete data g_h are fixed, the discrete operator acts as

$$\mathcal{P}_h(\cdot, g_h) : X_h \longrightarrow Y_h.$$

The numerical problem is

$$\mathcal{P}_h(u_h, g_h) = 0, \quad (1.2)$$

with $u_h \in X_h$. We first use the simplest notation, in which the exact solution can be inserted into the discrete equations and compared directly with u_h . The common situation in which sampling or interpolation is required is explained just after the basic error argument.

Definition 1.2 (Consistency). Let u be a sufficiently smooth exact solution, and let g_h denote the discrete representation of the exact data. The residual obtained by substituting u into the scheme is the *truncation error*

$$\tau_h = \mathcal{P}_h(u, g_h) \in Y_h.$$

The method is consistent if

$$\|\tau_h\|_{Y_h} \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Definition 1.3 (Stability). A family of discrete problems is stable if perturbations of the discrete data produce uniformly controlled perturbations of the discrete solution. For a linear operator $A_h : X_h \rightarrow Y_h$ and the problem $A_h u_h = f_h$, a standard form is

$$\|A_h^{-1}\|_{\mathcal{L}(Y_h, X_h)} \leq C \quad \text{with } C \text{ independent of } h.$$

Definition 1.4 (Convergence). A method is convergent if the numerical solution approaches the exact solution in the chosen solution norm:

$$\|u_h - u\|_{X_h} \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Proposition 1.5 (The basic error mechanism for linear problems). Suppose $A_h : X_h \rightarrow Y_h$, $A_h u_h = f_h$, and the exact solution can be inserted into the scheme so that $A_h u = f_h + \tau_h$. If $\|A_h^{-1}\| \leq C$ uniformly, then

$$\|u - u_h\|_{X_h} \leq C \|\tau_h\|_{Y_h}.$$

Thus consistency plus uniform stability implies convergence.

Proof. The error $e_h = u - u_h$ satisfies $A_h e_h = \tau_h$, so $e_h = A_h^{-1} \tau_h$. □

Sampling, interpolation, and the approximation property

The preceding notation deliberately suppresses a technical point. In many methods, u is a function whereas an element of X_h is a vector of nodal values, cell averages, or coefficients. The expression $A_h u$ and the difference $u_h - u$ are then not literally defined.

Introduce a sampling or interpolation operator

$$I_h : X \longrightarrow X_h.$$

The precise versions of consistency and discrete convergence are then

$$\|A_h I_h u - f_h\|_{Y_h} \longrightarrow 0, \quad \|u_h - I_h u\|_{X_h} \longrightarrow 0.$$

Stability gives the second statement from the first exactly as above. To conclude that the numerical method approximates the function u , one also requires an *approximation property*. If $X_h \subset X$, a typical assumption is

$$\|u - I_h u\|_X \longrightarrow 0.$$

With compatible norms, the triangle inequality then gives

$$\|u - u_h\|_X \leq \|u - I_h u\|_X + \|I_h u - u_h\|_X \longrightarrow 0.$$

If X_h consists only of grid vectors, use a reconstruction $R_h : X_h \rightarrow X$. The corresponding approximation property is $\|u - R_h I_h u\|_X \rightarrow 0$; if R_h is uniformly bounded, then

$$\|u - R_h u_h\|_X \leq \|u - R_h I_h u\|_X + C_R \|I_h u - u_h\|_{X_h} \longrightarrow 0.$$

Thus, in the fully precise formulation, approximation, consistency, and stability together imply convergence to the continuous solution.

Scope of the Lax equivalence theorem

The slogan *consistency plus stability implies convergence* is often associated with the Lax–Richtmyer theorem. That theorem is classically formulated for linear, well-posed initial-value problems and should not be quoted as an unrestricted theorem for every PDE discretization. This is in contrast with the Dahlquist Equivalence Theorem on numerical methods for ODEs.

Exercises

Exercise 1.1 (Absolute versus relative conditioning). Let $R(g) = g^{-1}$ for $g > 0$. Compute the absolute sensitivity $|R'(g)|$ and the relative condition number. Explain why the two quantities tell different stories near $g = 0$.

Exercise 1.2 (A scalar model). Consider $a_h u_h = f_h$ with $a_h > 0$, $a_h \rightarrow a > 0$, and $f_h \rightarrow f$. Identify a consistency error and a stability constant, and prove $u_h \rightarrow f/a$.

Exercise 1.3 (Residual versus error). Construct a 2×2 linear system for which a small residual does not imply a small error in the Euclidean norm. Relate the amplification to the condition number of the matrix.

Exercise 1.4 (Nonlinear extension). Suppose $F_h : X_h \rightarrow X_h$ is differentiable near a discrete solution and $F'_h(u_h)^{-1}$ is uniformly bounded. Use a first-order Taylor expansion to formulate a local analogue of the basic error mechanism.

2 Elliptic tools: Green identities and maximum principles

2.1 Gauss–Green formulas

Let $\Omega \subset \mathbb{R}^d$ be bounded with sufficiently regular boundary, and let n denote the outward unit normal. For $u \in C^1(\overline{\Omega})$, the divergence theorem gives

$$\int_{\Omega} \partial_i u \, dx = \int_{\partial\Omega} u n_i \, ds, \quad i = 1, \dots, d. \quad (2.1)$$

Consequently,

$$\int_{\Omega} \nabla u \, dx = \int_{\partial\Omega} u n \, ds, \quad (2.2)$$

$$\int_{\Omega} \Delta u \, dx = \int_{\partial\Omega} \partial_n u \, ds, \quad (2.3)$$

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = - \int_{\Omega} v \Delta u \, dx + \int_{\partial\Omega} v \partial_n u \, ds. \quad (2.4)$$

The last identity is Green’s first identity and is the bridge from strong to weak formulations.

2.2 Harmonic functions

A function $u \in C^2(\Omega)$ is harmonic if $\Delta u = 0$ in Ω . Harmonic functions satisfy the mean-value property:

$$u(x) = \int_{\partial B_r(x)} u \, ds = \int_{B_r(x)} u \, dy \quad \text{whenever } \overline{B_r(x)} \subset \Omega.$$

Conversely, a continuous function with the mean-value property is harmonic.

Theorem 2.1 (Maximum principles for harmonic functions). *Let Ω be connected and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic. Then*

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \quad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

If an extremum is attained at an interior point, then u is constant.

The strong statement follows from the mean-value property: if $u(x_0)$ equals the global maximum, then the average of u over every small ball about x_0 can equal that maximum only when u is constant on the ball. Connectedness propagates the conclusion.

Corollary 2.2 (Positivity). *If u is harmonic, $u = g \geq 0$ on $\partial\Omega$, and g is not identically zero, then $u > 0$ in Ω .*

2.3 Poisson’s equation: uniqueness and stability

Consider

$$-\Delta u = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega. \quad (2.5)$$

Corollary 2.3 (Uniqueness). *There is at most one classical solution of (2.5).*

Proof. The difference of two solutions is harmonic and vanishes on $\partial\Omega$; the maximum principle applied to w and $-w$ gives $w = 0$. $\square$

An equivalent energy proof uses (2.4):

$$0 = - \int_{\Omega} w \Delta w \, dx = \int_{\Omega} |\nabla w|^2 \, dx,$$

because $w = 0$ on the boundary.

Theorem 2.4 (A maximum-norm a priori estimate). *Suppose Ω is contained in a ball $B_R(x_0)$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ solves (2.5). Then*

$$\|u\|_{L^\infty(\Omega)} \leq \|g\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f\|_{L^\infty(\Omega)}.$$

Proof. Use the barrier $\phi(x) = (R^2 - |x - x_0|^2)/(2d)$, for which $-\Delta\phi = 1$ and $\phi \geq 0$ on Ω , and apply the maximum principle to suitable combinations of u , ϕ , and the boundary norm of g . $\square$

If $u, \tilde{u}$ correspond to data (f, g) and $(\tilde{f}, \tilde{g})$, the same estimate gives

$$\|u - \tilde{u}\|_{L^\infty(\Omega)} \leq \|g - \tilde{g}\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f - \tilde{f}\|_{L^\infty(\Omega)}.$$

This is continuous dependence, hence the stability part of well-posedness.

2.4 A general second-order operator

For clarity, write a linear operator in nondivergence form. Let $\Omega \subset \mathbb{R}^d$ be bounded and let

$$\mathcal{L}u = - \sum_{i,j=1}^d a_{ij}(x) \partial_{ij}u + \sum_{i=1}^d b_i(x) \partial_i u + c(x)u.$$

Assume that $A = (a_{ij})$ is symmetric and uniformly elliptic:

$$\lambda |\xi|^2 \leq \xi^\top A(x) \xi \leq \Lambda |\xi|^2, \quad \lambda > 0.$$

Theorem 2.5 (Weak maximum and comparison principles). *If $a_{ij}, b_i, c \in C(\overline{\Omega})$, with $c \geq 0$, then, for every $u \in C^2(\Omega) \cap C(\overline{\Omega})$:*

1. *if $\mathcal{L}u \leq 0$ in Ω , then*

$$\max_{\overline{\Omega}} u \leq \max \left\{ 0, \max_{\partial\Omega} u \right\};$$

if $c \equiv 0$, the zero may be omitted;

2. *if $\mathcal{L}u \leq \mathcal{L}v$ in Ω and $u \leq v$ on $\partial\Omega$, then $u \leq v$ in $\overline{\Omega}$.*

Remark 2.6. Analogous results hold for divergence-form operators $-\nabla \cdot (A \nabla u) + b \cdot \nabla u + cu$, but the natural weak hypotheses and proof are formulated in Sobolev spaces.

Exercises

Exercise 2.1 (Green's second identity). Derive

$$\int_{\Omega} (u \Delta v - v \Delta u) \, dx = \int_{\partial\Omega} (u \partial_n v - v \partial_n u) \, ds.$$

Exercise 2.2 (Maximum-principle uniqueness). Let $c \geq 0$. Prove uniqueness for $\mathcal{L}u = f$ with prescribed Dirichlet data by applying the comparison principle to the difference of two solutions.

Exercise 2.3 (Constructing a barrier). For $\Omega = (0, 1)^d$, construct a quadratic $\phi \geq 0$ satisfying $-\Delta\phi = 1$. Compare the resulting stability constant with the ball-based constant above.

Exercise 2.4 (Why $c \geq 0$?). Give a one-dimensional example showing that the maximum principle can fail when the zeroth-order coefficient is sufficiently negative.

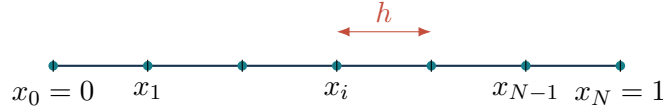
3 Centered finite differences in one dimension

3.1 The model problem and its discretization

Consider

$$-u'' = f \quad \text{in } (0, 1), \quad u(0) = u(1) = 0. \quad (3.1)$$

Choose $N \in \mathbb{N}$, set $h = 1/N$, and define $x_i = ih$, $i = 0, \dots, N$.



The centered second difference is

$$\delta_h^2 v(x) = \frac{v(x+h) - 2v(x) + v(x-h)}{h^2}.$$

The finite-difference scheme seeks nodal values $U_i \approx u(x_i)$ such that

$$(\mathcal{L}_h U)_i := -\delta_h^2 U_i = f_i := f(x_i), \quad i = 1, \dots, N-1, \quad U_0 = U_N = 0. \quad (3.2)$$

In matrix form,

$$A_h \mathbf{U} = \mathbf{F}, \quad A_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

The matrix is symmetric positive definite.

3.2 The discrete maximum principle

Theorem 3.1 (Discrete maximum principle). *Let $V = (V_0, \dots, V_N)$. If $(\mathcal{L}_h V)_i \leq 0$ for $i = 1, \dots, N-1$, then*

$$\max_{0 \leq i \leq N} V_i = \max \{V_0, V_N\}.$$

If an interior maximum is attained and $\mathcal{L}_h V \leq 0$, then the maximum propagates through neighboring nodes; hence V is constant.

Proof. At an interior maximizer n ,

$$0 \leq -h^2(\mathcal{L}_h V)_n = V_{n+1} - 2V_n + V_{n-1},$$

whereas maximality gives $(V_{n+1} + V_{n-1})/2 \leq V_n$. Equality is forced, so both neighbors also attain the maximum. Repetition reaches the boundary. $\square$

3.3 A sharp stability estimate

Set $w(x) = x(1 - x)$ and $W_i = w(x_i)$. Since w is quadratic,

$$\mathcal{L}_h W_i = -w''(x_i) = 2, \quad W_0 = W_N = 0, \quad 0 \leq W_i \leq \frac{1}{4}.$$

For an arbitrary grid function V , apply the discrete maximum principle to

$$Z_i^\pm = \pm V_i - \frac{1}{2} \|\mathcal{L}_h V\|_{\ell^\infty} W_i.$$

This yields:

Lemma 3.2 (Maximum-norm stability). *For every grid function V ,*

$$\|V\|_{\ell^\infty} \leq \max\{|V_0|, |V_N|\} + \frac{1}{8} \|\mathcal{L}_h V\|_{\ell^\infty}. \quad (3.3)$$

In particular, the homogeneous discrete problem has only the zero solution. Since the system is square and finite dimensional, uniqueness also implies existence.

3.4 Consistency and convergence

For $u \in C^4([0, 1])$, Taylor's theorem with remainder gives

$$u''(x) - \delta_h^2 u(x) = -\frac{h^2}{24} \left(u^{(4)}(\xi_-) + u^{(4)}(\xi_+) \right), \quad (3.4)$$

where $\xi_- \in (x - h, x)$ and $\xi_+ \in (x, x + h)$. Therefore

$$\left| u''(x) - \delta_h^2 u(x) \right| \leq \frac{h^2}{12} \|u^{(4)}\|_{L^\infty(0,1)}.$$

Note that the remainder in (3.4) carries a minus sign when written as $u'' - \delta_h^2 u$. The sign does not affect the norm bound, but it matters in a pointwise asymptotic expansion.

Let $e_i = u(x_i) - U_i$. Since $e_0 = e_N = 0$,

$$(\mathcal{L}_h e)_i = (\mathcal{L}_h I_h u)_i - f_i = u''(x_i) - \delta_h^2 u(x_i).$$

Combining the truncation bound with (3.3) proves:

Theorem 3.3 (Second-order maximum-norm convergence). *If $u \in C^4([0, 1])$ solves (3.1) and U solves (3.2), then*

$$\max_{0 \leq i \leq N} |u(x_i) - U_i| \leq \frac{h^2}{96} \|u^{(4)}\|_{L^\infty(0,1)}. \quad (3.5)$$

The reusable proof pattern

$$\begin{array}{c} \boxed{\text{error equation}} + \boxed{\text{uniform stability}} \\ + \boxed{\text{truncation estimate}} \implies \boxed{\text{convergence estimate}}. \end{array}$$

3.5 Conditioning and solver cost

Spectral estimate for the Poisson matrix

The eigenvalues of A_h are

$$\lambda_k = \frac{4}{h^2} \sin^2 \left(\frac{k\pi}{2N} \right), \quad k = 1, \dots, N-1.$$

Hence

$$\text{cond}_2(A_h) \sim \frac{4}{\pi^2 h^2}.$$

Refinement improves approximation but worsens algebraic conditioning. This observation motivates preconditioners and multigrid methods.

Exercises

Exercise 3.1 (Exactness on quadratics). Prove that $\delta_h^2 p = p''$ for every polynomial of degree at most three. Find the error explicitly for $p(x) = x^4$.

Exercise 3.2 (A nonzero boundary condition). Derive the matrix system for $u(0) = \alpha$, $u(1) = \beta$. Show how the boundary values enter only the first and last entries of the right-hand side.

Exercise 3.3 (Energy stability). Multiply $(\mathcal{L}_h U)_i = f_i$ by hU_i , sum over the interior nodes, and prove a discrete integration-by-parts identity. Deduce a stability estimate in the discrete H^1 seminorm.

Exercise 3.4 (Observed order). Use the exact solution $u(x) = \sin(\pi x)$. Compute the discrete solution for successively halved h , and verify that $\|I_h u - U\|_{\ell^\infty}$ decreases by a factor close to four.

4 Weak formulations and energy principles

4.1 Why the strong formulation is too restrictive

The classical problem $-u'' = f$ appears to require two derivatives of u and pointwise values of f . Many physically natural loads do not produce such smooth solutions.

Example 4.1 (A discontinuous load). If $f = \chi_{[0.4, 0.6]}$, then the solution of $-u'' = f$, $u(0) = u(1) = 0$, is positive and piecewise quadratic. It is C^1 , but its second derivative jumps at 0.4 and 0.6.

Example 4.2 (A point load). Let $f = -\delta_{1/2}$. The weak solution is the negative tent function

$$u(x) = \begin{cases} -x/2, & 0 \leq x \leq 1/2, \\ -(1-x)/2, & 1/2 \leq x \leq 1. \end{cases}$$

It belongs to $H_0^1(0, 1)$, but it is not classically differentiable at $x = 1/2$.

With the convention $-u'' = f$, a nonnegative transverse load produces a nonnegative displacement for homogeneous Dirichlet data. A negative tent corresponds to the negative point load $-\delta_{1/2}$, not to a positive load.

4.2 Sobolev spaces

Define

$$L^2(0, 1) = \left\{ v : \|v\|_{L^2}^2 = \int_0^1 |v|^2 \, dx < \infty \right\},$$

$$H^1(0, 1) = \left\{ v \in L^2(0, 1) : v' \in L^2(0, 1) \right\}, \quad H_0^1(0, 1) = \overline{C_c^\infty(0, 1)}^{H^1}.$$

In one dimension, functions in H^1 have continuous representatives, and

$$H_0^1(0, 1) = \left\{ v \in H^1(0, 1) : v(0) = v(1) = 0 \right\}.$$

Poincaré's inequality gives

$$\|v\|_{L^2(0, 1)} \leq C_P \|v'\|_{L^2(0, 1)} \quad \forall v \in H_0^1(0, 1).$$

Thus

$$\|v\|_V := \|v'\|_{L^2(0, 1)}$$

is a norm on $V = H_0^1(0, 1)$, equivalent to the full H^1 norm. The dual space $V' = H^{-1}(0, 1)$ contains $L^2(0, 1)$ and, in one dimension, point loads.

Norm versus seminorm

The quantity $\|v'\|_{L^2}$ is only a seminorm on $H^1(0, 1)$, because constants have zero derivative. It becomes a genuine norm on $H_0^1(0, 1)$ by Poincaré's inequality.

4.3 Derivation of the weak problem

Multiply $-u'' = f$ by a test function v satisfying $v(0) = v(1) = 0$, integrate, and use integration by parts:

$$-\int_0^1 u'' v \, dx = \int_0^1 u' v' \, dx = \langle f, v \rangle_{V', V}.$$

The weak formulation is:

$$\text{Find } u \in V \text{ such that } \quad a(u, v) = F(v) \quad \forall v \in V, \quad (4.1)$$

with

$$a(u, v) = \int_0^1 u' v' \, dx, \quad F(v) = \langle f, v \rangle_{V', V}.$$

Theorem 4.3 (Well-posedness of the weak problem). *For every $F \in V'$, problem (4.1) has a unique solution, and*

$$\|u\|_V \leq \|F\|_{V'}.$$

Proof. The bilinear form is continuous and coercive:

$$|a(w, v)| \leq \|w\|_V \|v\|_V, \quad a(v, v) = \|v\|_V^2.$$

Hence, the Lax–Milgram Lemma (see later) gives existence and uniqueness. Testing with $v = u$ gives $\|u\|_V^2 = F(u) \leq \|F\|_{V'} \|u\|_V$. $\square$

4.4 Dirichlet's principle

Define the potential energy

$$J(v) = \frac{1}{2} a(v, v) - F(v) = \frac{1}{2} \int_0^1 |v'|^2 \, dx - F(v). \quad (4.2)$$

Theorem 4.4 (Weak solution equals energy minimizer). *The weak solution u of (4.1) is the unique minimizer of J over V .*

Proof. For $w \in V$,

$$J(u + w) = J(u) + a(u, w) - F(w) + \frac{1}{2} a(w, w) = J(u) + \frac{1}{2} a(w, w) \geq J(u).$$

Conversely, if u minimizes J , then the derivative of $t \mapsto J(u + tw)$ at $t = 0$ vanishes for every $w \in V$, which gives $a(u, w) = F(w)$. $\square$

Nonhomogeneous Dirichlet data

If $u(0) = \alpha$ and $u(1) = \beta$, choose a lifting $G \in H^1(0, 1)$ with the prescribed traces and write $u = G + w$, $w \in H_0^1(0, 1)$. The admissible set is the affine space $G + H_0^1(0, 1)$; variations still belong to $H_0^1(0, 1)$.

Exercises

Exercise 4.1 (Poincaré inequality in one dimension). Use $v(x) = \int_0^x v'(s) \, ds$ and Cauchy–Schwarz to prove a Poincaré inequality on $H_0^1(0, 1)$. Find a valid constant.

Exercise 4.2 (Point evaluation). Prove that $|v(1/2)| \leq C \|v'\|_{L^2}$ for $v \in H_0^1(0, 1)$. Conclude that $\delta_{1/2} \in H^{-1}(0, 1)$.

Exercise 4.3 (Energy of the tent function). Verify distributionally that the tent function in the point-load example solves $-u'' = -\delta_{1/2}$. Compute $J(u)$.

Exercise 4.4 (Reaction term). Derive the weak and energy formulations for $-u'' + cu = f$, $c \geq 0$. Identify the coercivity constant in the H_0^1 norm used above.

5 Finite elements and finite volumes in one dimension

5.1 The conforming Galerkin idea

The finite element method restricts the weak problem to a finite-dimensional subspace $V_h \subset V$:

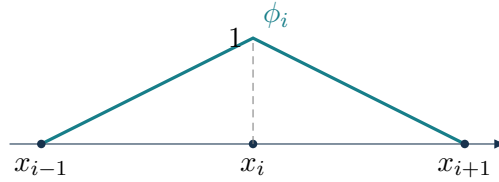
$$\text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h. \quad (5.1)$$

On the grid from section 3, choose the continuous, piecewise affine space

$$V_h = \left\{ v_h \in H_0^1(0, 1) : v_h|_{[x_{i-1}, x_i]} \in \mathbb{P}_1 \text{ for } i = 1, \dots, N \right\}.$$

Its nodal basis consists of hat functions ϕ_i , $i = 1, \dots, N-1$, satisfying $\phi_i(x_j) = \delta_{ij}$:

$$\phi_i(x) = \begin{cases} (x - x_{i-1})/h, & x \in [x_{i-1}, x_i], \\ (x_{i+1} - x)/h, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$



Writing

$$u_h = \sum_{j=1}^{N-1} U_j \phi_j$$

and testing with every ϕ_i gives

$$K\mathbf{U} = \mathbf{b}, \quad K_{ij} = \int_0^1 \phi_j' \phi_i' \, dx, \quad b_i = F(\phi_i). \quad (5.2)$$

For a uniform grid,

$$K = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

5.2 Well-posedness and error analysis

Because $V_h \subset V$, the same continuity and coercivity constants apply to the discrete problem. Hence (5.1) has a unique solution. Subtracting the continuous and discrete equations gives *Galerkin orthogonality*:

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h. \quad (5.3)$$

Theorem 5.1 (Céa's lemma). *If a is continuous with constant M and coercive with constant α , then*

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Proof. For any $v_h \in V_h$, coercivity and (5.3) give

$$\alpha \|u - u_h\|_V^2 \leq a(u - u_h, u - u_h) = a(u - u_h, u - v_h) \leq M \|u - u_h\|_V \|u - v_h\|_V.$$

□

For piecewise affine elements and $u \in H^2(0, 1)$, interpolation yields

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1} \leq Ch \|u\|_{H^2}.$$

Thus the energy error is first order. A duality argument gives a second-order L^2 estimate under the corresponding elliptic regularity.

5.3 When finite elements reproduce finite differences

For $F(v) = \int_0^1 f v \, dx$, the i -th finite element equation is

$$-\frac{1}{h}U_{i-1} + \frac{2}{h}U_i - \frac{1}{h}U_{i+1} = \int_{x_{i-1}}^{x_{i+1}} f \phi_i \, dx.$$

Approximating the right-hand side by the nodal quadrature

$$\int_{x_{i-1}}^{x_{i+1}} f \phi_i \, dx \approx h f(x_i)$$

and dividing by h recovers the centered finite-difference equation.

The equivalence uses quadrature

The linear finite element stiffness matrix and the centered finite-difference matrix have the same tridiagonal pattern, up to a row scaling. Exact equivalence for a general load requires a specific approximation of the finite element load vector; it is not an identity with exact integration.

5.4 A finite volume derivation

Let the control volume about x_i be $I_i = [x_{i-1/2}, x_{i+1/2}]$. Integrating $-u'' = f$ over I_i gives the exact balance

$$-u'(x_{i+1/2}) + u'(x_{i-1/2}) = \int_{I_i} f \, dx. \quad (5.4)$$

Approximate the face fluxes by

$$u'(x_{i+1/2}) \approx \frac{U_{i+1} - U_i}{h}, \quad u'(x_{i-1/2}) \approx \frac{U_i - U_{i-1}}{h},$$

and use the midpoint rule on the right. After division by h ,

$$\frac{-U_{i+1} + 2U_i - U_{i-1}}{h^2} = f(x_i).$$

Thus the same algebraic scheme arises from three perspectives: pointwise differentiation, a variational principle with nodal quadrature, and local flux balance.

By construction, the finite volume approach allows to show a local conservation property as follows. Write the balance on cell i as

$$-q_{i+1/2} + q_{i-1/2} = \int_{I_i} f \, dx, \quad q_{i+1/2} \approx \frac{U_{i+1} - U_i}{h}.$$

Summing the finite volume equations over a consecutive block of control volumes, that is from $i = m$ to $i = n$, with $m < n$, gives

$$\sum_{i=m}^n (-q_{i+1/2} + q_{i-1/2}) = q_{m-1/2} - q_{n+1/2} = \int_{x_{m-1/2}}^{x_{n+1/2}} f \, dx.$$

Every internal face flux appears once with each sign and cancels. The remaining equation says that the net outward flux through the boundary of the block equals the total source inside it. This is local conservation on any union of control volumes.

How the viewpoints differ

- Finite differences make truncation and maximum-principle arguments transparent.
- Finite elements naturally handle weak solutions, complex geometry, and energy-norm error estimates.
- Finite volumes enforce a local conservation law and are often preferred when flux balance is central.

The algebraic coincidence in this one-dimensional example should not be mistaken for complete equivalence of the methods.

Exercises

Exercise 5.1 (Element assembly). Compute the 2×2 stiffness matrix on a single interval $[x_{i-1}, x_i]$, then assemble the global tridiagonal matrix.

Exercise 5.2 (Galerkin best approximation). For the Poisson bilinear form, show that u_h is the orthogonal projection of u onto V_h in the energy inner product. Improve Céa's estimate to an equality involving the best approximation error.

Exercise 5.3 (A point load in FEM). Take $F(v) = -v(1/2)$ on a mesh containing $1/2$. Write the finite element load vector and show that the exact tent solution belongs to V_h . What does the discrete solution equal?